KAIZEN MATHS

Lesson starter

Inverse Trig Differentiation

Tangents and Normals

Name _____

S1

Find the tangent to $y = \csc^{-1}(-4x - 5)$ at $x = -2$.

S2

Find the normal to $y = \tan^{-1}(-7x + 20)$ at $x = 3$.

S3

Find the tangent to $y = \csc^{-1}(-3x - 11)$ at $x = -3$.

S4

Find the normal to $y = \sec^{-1}(7x + 19)$ at $x = -3$.

Show your working

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ALG-893C-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Inverse Trig Differentiation

Tangents and Normals

Name _____

E1

Find the tangent to $y = \sec^{-1}(-5x - 13)$ at $x = -3$.

E2

Find the normal to $y = \tan^{-1}(-3x - 8)$ at $x = -3$.

E3

Find the tangent to $y = \tan^{-1}(-9x + 16)$ at $x = 2$.

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Show your working

Teacher answers & working

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S1

Find the tangent to $y = \csc^{-1}(-4x - 5)$ at $x = -2$.

Answer:

$$y - \csc^{-1}(3) = \frac{4}{|3|\sqrt{(3)^2 - 1}}(x + 2)$$

Working:

$u(-2) = -4(-2) + (-5)$

$u(-2) = 3$

$\frac{dy}{du} = -\frac{1}{|u|\sqrt{u^2 - 1}}$

$\frac{du}{dx} = -4$

$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$

$m = (-\frac{1}{|3|\sqrt{(3)^2 - 1}})(-4)$

The tangent gradient is $m = \frac{4}{|3|\sqrt{(3)^2 - 1}}$.

The point on the curve is $(-2, \csc^{-1}(3))$.

Use point-gradient form: $y - y_1 = m(x - x_1)$.

$$y - \csc^{-1}(3) = \frac{4}{|3|\sqrt{(3)^2 - 1}}(x + 2)$$

Teacher answers & working

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S2

Find the normal to $y = \tan^{-1}(-7x + 20)$ at $x = 3$.

Answer:

$$y - \tan^{-1}(-1) = \frac{1 + (-1)^2}{7}(x - 3)$$

Working:

$u(3) = -7(3) + (20)$

$u(3) = -1$

$\frac{dy}{du} = \frac{1}{1 + u^2}$

$\frac{du}{dx} = -7$

$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$

$m = (\frac{1}{1 + (-1)^2})(-7)$

The tangent gradient is $m_{\tan} = -\frac{7}{1 + (-1)^2}$.

The normal gradient is the negative reciprocal: $m_{\text{norm}} = -1/m_{\tan} = \frac{1 + (-1)^2}{7}$.

The point on the curve is $(3, \tan^{-1}(-1))$.

Using point-gradient form, the normal is $y - \tan^{-1}(-1) = \frac{1 + (-1)^2}{7}(x - 3)$.

Teacher answers & working

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S3

Find the tangent to $y = \csc^{-1}(-3x - 11)$ at $x = -3$.

Answer:

$$y - \csc^{-1}(-2) = \frac{3}{|-2|\sqrt{(-2)^2 - 1}}(x + 3)$$

Working:

$u(-3) = -3(-3) + (-11)$

$u(-3) = -2$

$\frac{dy}{du} = -\frac{1}{|u|\sqrt{u^2 - 1}}$

$\frac{du}{dx} = -3$

$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$

$m = (-\frac{1}{|-2|\sqrt{(-2)^2 - 1}})(-3)$

The tangent gradient is $m = \frac{3}{|-2|\sqrt{(-2)^2 - 1}}$.

The point on the curve is $(-3, \csc^{-1}(-2))$.

Use point-gradient form: $y - y_1 = m(x - x_1)$.

$$y - \csc^{-1}(-2) = \frac{3}{|-2|\sqrt{(-2)^2 - 1}}(x + 3)$$

Teacher answers & working

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S4

Find the normal to $y = \sec^{-1}(7x + 19)$ at $x = -3$.

Answer:

$$y - \sec^{-1}(-2) = -\frac{|-2|\sqrt{(-2)^2 - 1}}{7}(x + 3)$$

Working:

$u(-3) = 7(-3) + (19)$

$u(-3) = -2$

$\frac{dy}{du} = \frac{1}{|u|\sqrt{u^2 - 1}}$

$\frac{du}{dx} = 7$

$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$

$m = (\frac{1}{|-2|\sqrt{(-2)^2 - 1}})(7)$

The tangent gradient is $m_{\tan} = \frac{7}{|-2|\sqrt{(-2)^2 - 1}}$.

The normal gradient is the negative reciprocal: $m_{\text{norm}} = -1/m_{\tan} = -\frac{|-2|\sqrt{(-2)^2 - 1}}{7}$.

The point on the curve is $(-3, \sec^{-1}(-2))$.

Using point-gradient form, the normal is $y - \sec^{-1}(-2) = -\frac{|-2|\sqrt{(-2)^2 - 1}}{7}(x + 3)$.

Teacher answers & working

Inverse Trig Differentiation · Tangents and Normals

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E1

Find the tangent to $y = \sec^{-1}(-5x - 13)$ at $x = -3$.

Answer:

$$y - \sec^{-1}(2) = -\frac{5}{|2|\sqrt{(2)^2 - 1}}(x + 3)$$

Working:

$u(-3) = -5(-3) + (-13)$

$u(-3) = 2$

$\frac{dy}{du} = \frac{1}{|u|\sqrt{u^2 - 1}}$

$\frac{du}{dx} = -5$

$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$

$m = (\frac{1}{|2|\sqrt{(2)^2 - 1}})(-5)$

The tangent gradient is $m = -\frac{5}{|2|\sqrt{(2)^2 - 1}}$.

The point on the curve is $(-3, \sec^{-1}(2))$.

Use point-gradient form: $y - y_1 = m(x - x_1)$.

$$y - \sec^{-1}(2) = -\frac{5}{|2|\sqrt{(2)^2 - 1}}(x + 3)$$

Teacher answers & working

Inverse Trig Differentiation · Tangents and Normals

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E2

Find the normal to $y = \tan^{-1}(-3x - 8)$ at $x = -3$.

Answer:

$$y - \tan^{-1}(1) = \frac{1 + (1)^2}{3}(x + 3)$$

Working:

$$u(-3) = -3(-3) + (-8)$$

$$u(-3) = 1$$

$$\frac{dy}{du} = \frac{1}{1 + u^2}$$

$$\frac{du}{dx} = -3$$

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

$$m = (\frac{1}{1 + (1)^2})(-3)$$

The tangent gradient is $m_{\tan} = -\frac{3}{1 + (1)^2}$.

The normal gradient is the negative reciprocal: $m_{\text{norm}} = -1/m_{\tan} = \frac{1 + (1)^2}{3}$.

The point on the curve is $(-3, \tan^{-1}(1))$.

Using point-gradient form, the normal is $y - \tan^{-1}(1) = \frac{1 + (1)^2}{3}(x + 3)$.

Teacher answers & working

Inverse Trig Differentiation · Tangents and Normals

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E3

Find the tangent to $y = \tan^{-1}(-9x + 16)$ at $x = 2$.

Answer:

$$y - \tan^{-1}(-2) = -\frac{9}{1 + (-2)^2}(x - 2)$$

Working:

$u(2) = -9(2) + (16)$

$u(2) = -2$

$\frac{dy}{du} = \frac{1}{1 + u^2}$

$\frac{du}{dx} = -9$

$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$

$m = (\frac{1}{1 + (-2)^2})(-9)$

The tangent gradient is $m = -\frac{9}{1 + (-2)^2}$.

The point on the curve is $(2, \tan^{-1}(-2))$.

Use point-gradient form: $y - y_1 = m(x - x_1)$.

$$y - \tan^{-1}(-2) = -\frac{9}{1 + (-2)^2}(x - 2)$$